

Lumino

A companion to guide your child to sleep

Spring26-ENP64-Final Assignment

Ginny Dong & Kylinn Ding

Table of Contents

I. Product Description **2**

II. Target User **2**

III. Initial Design Research **3**

CLIMBER Analysis 3

Questionnaire 4

Task Analysis 4

IV. Lumino APP **6**

V. Lumino Physical Product **10**

Biomechanical Considerations 10

Anthropometrics Considerations 10

Musculoskeletal Injury Concerns 11

VII. App Interface Redesign **14**

VIII. Discussion and reflection **15**

IX. Appendix **16**

Questionnaire content 16

Questionnaire results 18

Task analysis 22

Usability testing script 25

I. Product Description

Lumino is a mushroom-shaped nightlight companion and app designed for young children ages 1–5 and their parents.

The physical component of the product is a nightlight modeled as a mushroom. Its diffuser top emits a slow, pulsing glow that subconsciously guides young children through a calming breathing rhythm, encouraging them to slow their breath and drift into sleep without direct parental intervention. The device features three primary interactions: a breathing light mode that cycles through a soothing inhale-exhale pattern, a story button that triggers pre-loaded bedtime stories or custom audios directly from the device, and an intercom button that allows children to call out to a parent without crying or leaving their bed. Each button is distinctly shaped and tactilely differentiated so that toddlers can operate them independently in the dark. The mushroom top of the device is made from a soft silicone material to prevent hurt in the dark, and is sized to sit comfortably on a nightstand or be held by small hands.

The product's digital component is a companion app intended primarily for parents. It allows caregivers to customize the breathing light's speed and color, upload or select bedtime stories and audio content, play sounds when the child activates the call button, and track sleep patterns over time.

Ultimately, the purpose of Lumino is to give young children a tangible, comforting anchor at bedtime. It provides the security of a parent's voice on demand, and reduces the dependency on adult presence to fall asleep. Products that address childhood sleep struggles tend to focus on the parent's experience; Lumino centers the child's, putting the tools for calm directly in their hands.

II. Target User

Children ages 1–5 are the primary target demographic for Lumino. This age group commonly experiences separation anxiety at bedtime, difficulty self-soothing after nighttime wake-ups, and overstimulation from transitioning between active play and rest. Unlike older children, toddlers cannot yet follow complex verbal instructions or engage meaningfully with screen-based sleep aids — they need a physical, sensory experience suited to their developmental level. Lumino

addresses this through interactions that are immediate and intuitive, requiring no reading, fine motor precision, or prior experience with technology.

Every physical design decision prioritizes safety, simplicity, and sensory comfort for this age group. The silicone material is soft, chew-safe, and drop-resistant. The buttons are large, tactilely distinct, and operable in the dark by small hands. The light is slow, warm, and non-stimulating.

Parents and caregivers of children in this age group are the secondary user. They are responsible for purchasing, setting up, and customizing Lumino, and they are directly affected by their child's sleep difficulties — including repeated nighttime check-ins and disrupted sleep. The companion app is designed with parents in mind, providing customization and control without adding complexity to the child-facing device. The intercom feature serves both users: it gives the child a sense of security and allows the parent to respond without physically entering the room.

III. Initial Design Research

CLIMBER Analysis

Using the CLIMBER framework, the target customer was identified as parents of toddlers ages 1–5 who are transitioning their child from co-sleeping or parent-accompanied sleep to sleeping independently. These parents want to establish a consistent bedtime routine that does not require their physical presence, while still maintaining a sense of connection and reassurance for their child. Their primary concern is whether their child is truly ready to self-soothe through the night and whether they are safe sleeping alone.

Mapping the life context revealed that this transition period is one of the more stressful early parenting milestones. Children's bedrooms lack any comforting sensory anchor that can substitute for a parent's presence once the door closes, and children in this stage have no independent coping mechanism to fall back on when they wake in the middle of the night. The core insight drawn from this analysis is that children making this transition do not just need light or sound — they need a tangible sense of parental presence, and parents need a way to remain responsive without physically re-entering the room. This directly informed Lumino's core mechanism: a breathing-rhythm light combined with on-demand intercom and story playback, all activated through simple one-touch interactions a toddler can operate independently.

Questionnaire

Nine responses were collected for the initial questionnaire, with the full list of questions and responses provided in the appendix. Most of the children represented in the sample were between the ages of 3–4, and a large majority of caregivers indicated that their child is currently transitioning to sleeping independently. Despite this, independent sleep remains a challenge for many children. Most respondents reported that their child wakes up during the night at least once, and many indicated that their child sometimes or often has difficulty falling asleep independently.

Caregivers also reported frequent involvement during bedtime routines. The majority indicated that they re-enter their child's room one to two times per night, while others reported returning even more frequently. When children call out at night, responses varied, with many caregivers choosing to respond verbally from another room, while others immediately went to their child. Bedtime was commonly described as a moderately stressful experience, with relatively few respondents reporting no stress.

In terms of current strategies, caregivers reported using approaches such as storytelling, singing, or talking to help their child fall asleep. Notably, most children were not using any interactive sleep aids. However, there was general openness toward potential solutions, as most respondents expressed neutral to positive attitudes toward a device that would allow children to communicate with caregivers without leaving bed. Overall, these findings suggest that both children and caregivers experience ongoing challenges during the transition to independent sleep, highlighting a need for solutions that support self-soothing while maintaining a sense of parental presence.

Task Analysis

A task analysis was conducted to better understand the physical and mental demands placed on both the child and the parent during the bedtime independence transition. The full analysis is attached in Appendix.

For the child, physical tasks include getting into bed, pulling up a blanket, reaching for a stuffed animal, calling out or crying, and often getting out of bed to find a parent before eventually settling. The mental demands are more significant: the child must process the absence of the

parent, manage fear or uncertainty about being alone in the dark, attempt to shift attention away from discomfort, and cycle between near-sleep and startling awake while listening for sounds outside the room.

For the parent, physical tasks include completing the bedtime routine, reading a story, tucking the child in, leaving the room, and re-entering to soothe when the child cries — often repeating this cycle multiple times before the child is fully asleep. Mentally, the parent manages guilt about leaving, uncertainty about whether the child is ready, and the ongoing decision of whether to return or hold firm on the routine.

This analysis highlighted that both users are actively struggling during the same event. Lumino's design addresses both sides: the breathing light and story button give the child a sensory anchor and a distraction from discomfort, while the intercom gives the parent a way to respond without disrupting the routine.

Conclusion

Overall, the findings highlight that the transition to independent sleep is a challenging process for both children and caregivers, characterized by frequent nighttime awakenings, ongoing parental intervention, and moderate levels of stress. Children often struggle with self-soothing and require reassurance, while caregivers attempt to balance responsiveness with encouraging independence. These insights reinforce the need for a solution that supports both users simultaneously. Lumino addresses this need by providing children with a sensory-based tool for calming and self-regulation through its breathing light and audio features, while also maintaining a sense of parental presence through its intercom functionality. By combining intuitive physical interaction with caregiver support through a companion app, Lumino offers a system designed to reduce bedtime stress and facilitate a smoother transition to independent sleep.

IV. Lumino APP

Interface design considerations

The Lumino companion app is designed to reduce cognitive load and support quick interaction during bedtime routines. The experience begins with a simple login and registration flow, followed by an account setup process that collects essential information such as the child's name, parent's name, email address, and password. This ensures a straightforward onboarding experience without unnecessary complexity.

The interface is organized into four main sections, including Home, Story, Intercom, and Profile, each corresponding to a core function of the system. On the Home screen, key controls such as brightness, volume, and sleep timer are presented using large sliders alongside a visual representation of the lamp, allowing caregivers to quickly understand and adjust the device state. The Story section supports content management through a playlist and a content library, where users can select preexisting audio or upload custom content from their device.

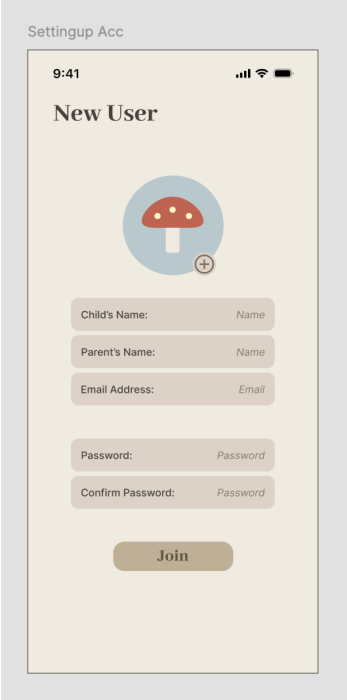
The Intercom screen focuses on immediacy and ease of use, featuring a large, clearly visible hold-to-talk button that enables direct communication with the child. The Profile section provides an overview of the parent and child account, along with usage history displayed through a calendar interface that tracks story playback and intercom activity over time. Additional system and privacy settings are also accessible within this section.

A warm color palette, consistent layout, and large touch targets improve usability in low-light conditions and support ease of use. Overall, the interface prioritizes clarity, efficiency, and emotional reassurance to support smoother bedtime routines.

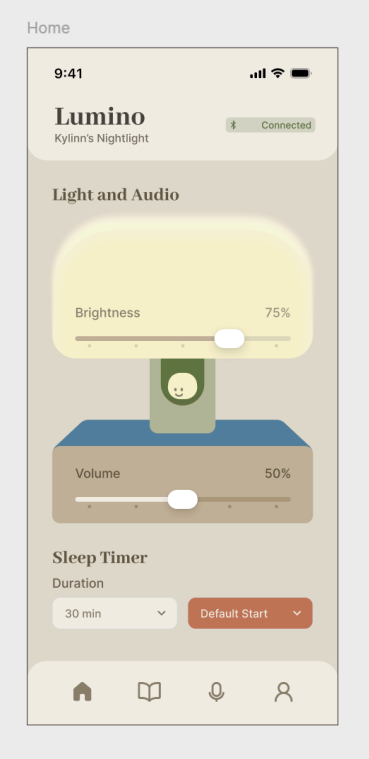
Login screen



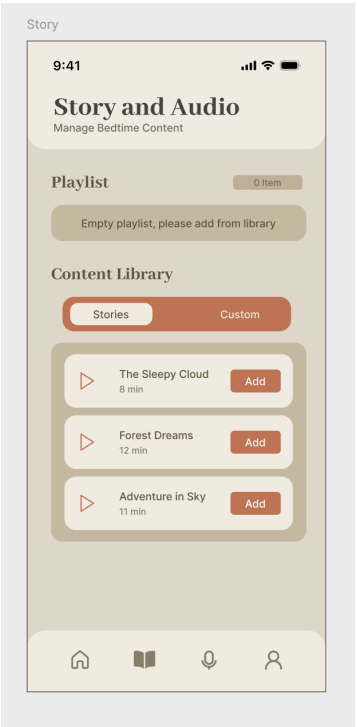
Account setup

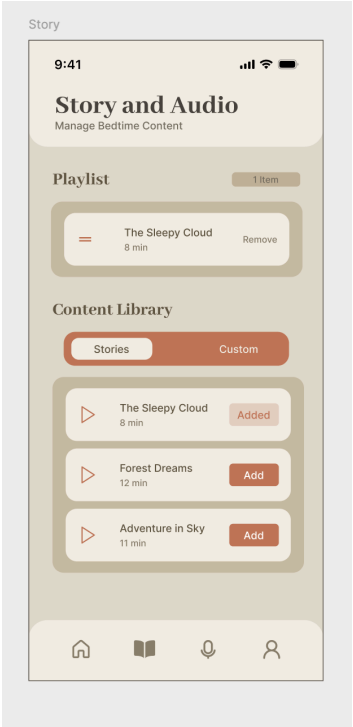


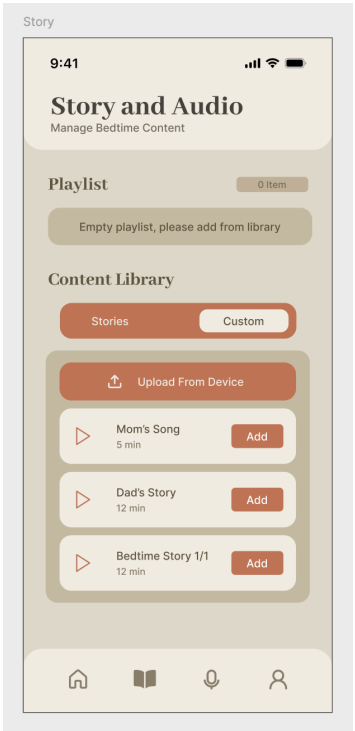
Homepage



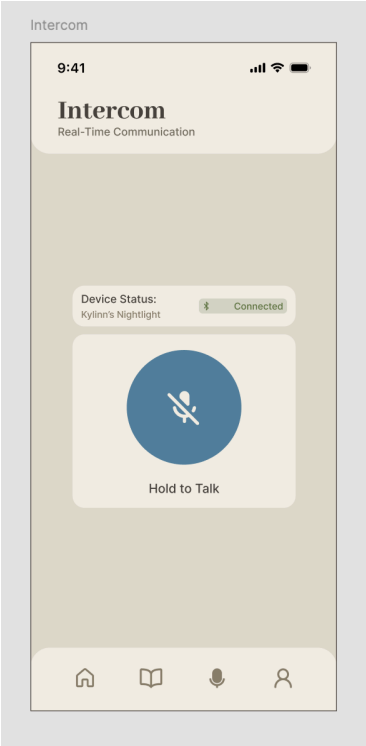
Story and audio options



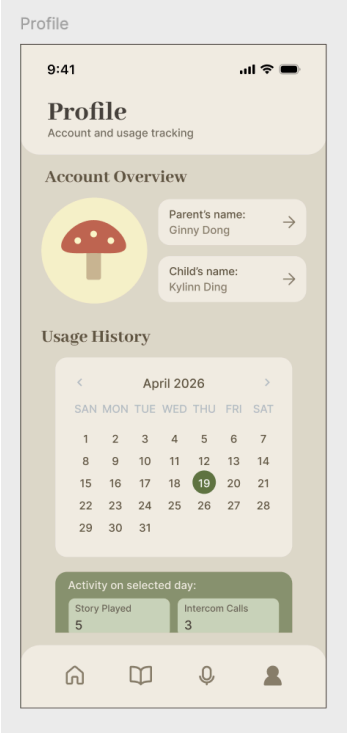


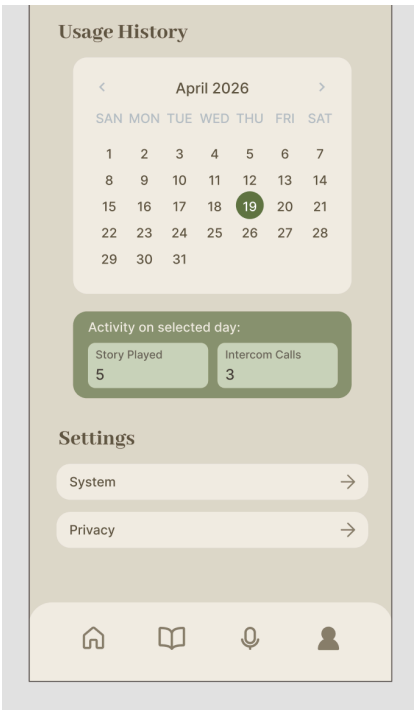


Intercom



Profile





V. Lumino Physical Product

Biomechanical Considerations

The tap-to-activate silicone top is designed to suit toddlers' limited fine motor control. An open-palm pat is instinctive and effortless for this age group, requiring no pinching, twisting, or precise finger placement. The two side buttons are sized for small fingers and include enough actuation resistance to prevent accidental activation during sleep, while still being pressable by a child acting intentionally.

Anthropometrics Considerations

Lumino is designed for children ages 1–5, whose hand dimensions and motor abilities differ significantly from those of adults. Interaction areas are sized based on pediatric anthropometric data to ensure accessibility without requiring precise control.

For palm-based interactions, young children typically have a palm width ranging approximately from 2 to 3 inches. As a result, the primary activation surface is designed within a smaller but reachable range that supports partial palm contact, allowing children to activate the device through an open-hand pat without needing full hand coverage or precise targeting.

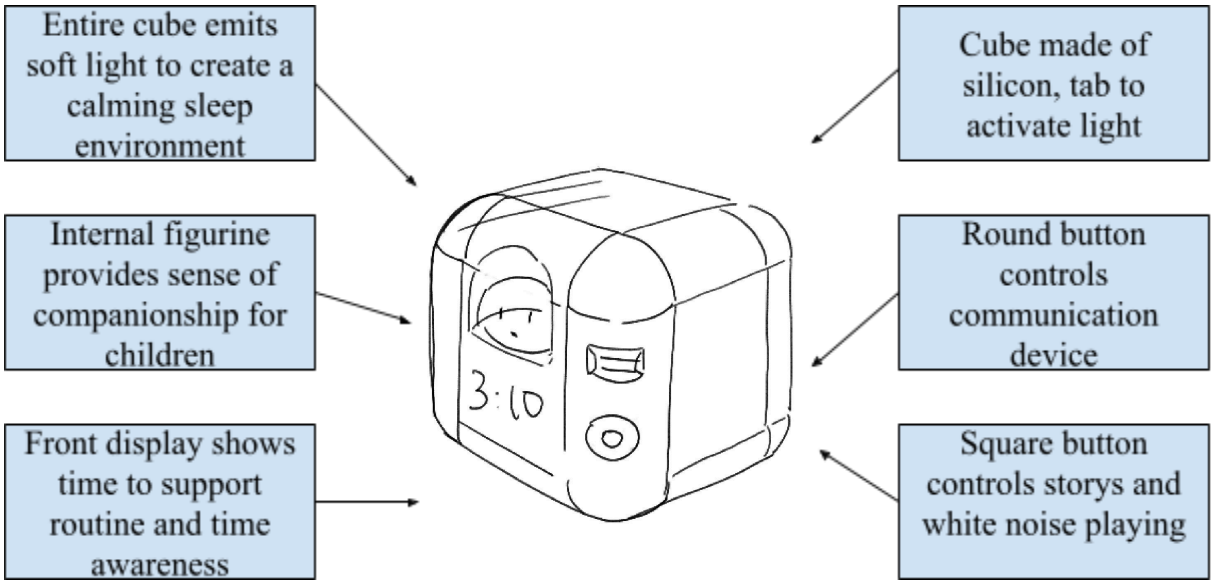
For finger-based inputs, toddler fingertip breadth generally falls between approximately 0.4 and 0.6 inches. Based on this, interactive elements are designed within a range that includes both larger, easily pressable controls and smaller, more constrained controls. Larger buttons fall within a range that exceeds fingertip width to support accurate pressing, while smaller buttons remain within a reduced size range to prevent accidental activation, relying on placement and tactile differentiation for usability.

Grip-related dimensions are informed adult users, as parents are mostly responsible for moving the device around. Adult hand breadth and grip comfort generally fall within a range of approximately 2.5 to 3 inches. Therefore, handling areas are designed within this range to ensure that caregivers can comfortably grasp and reposition the device.

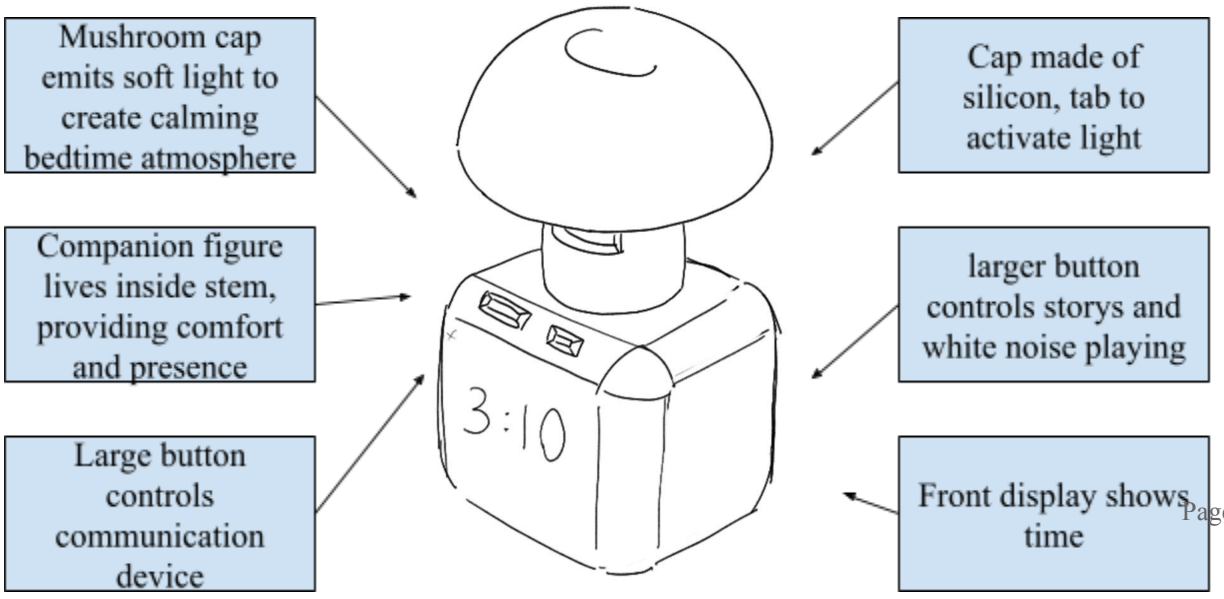
Musculoskeletal Injury Concerns

Lumino presents minimal musculoskeletal injury risk given its passive, stationary use case. However, several design decisions were made to address safety concerns specific to young children. Rounded corners on the base eliminate sharp edges that could cause injury if a child rolls into the device during sleep. The silicone top is soft, non-toxic, and bite-safe, which is critical for children under 3 who frequently mouth objects. A weighted base reduces tip-over risk, and cord-free magnetic charging removes any strangulation hazard entirely.

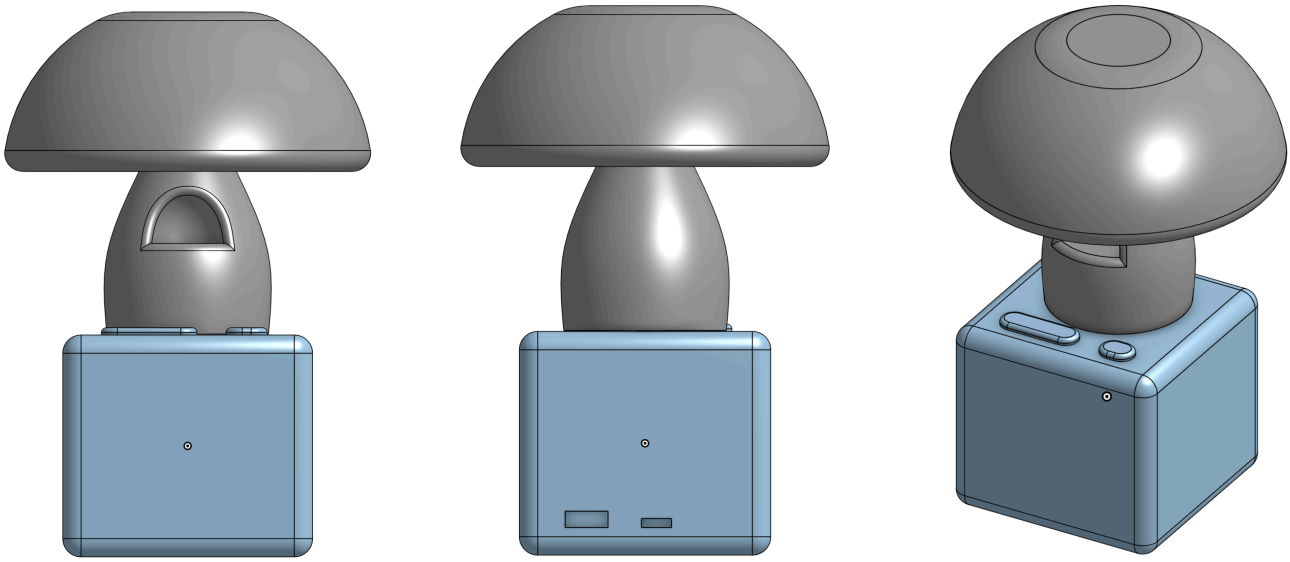
Iteration 1:



Iteration 2:



Iteration 2 finalized - CAD drawing:



Iteration finalized - rendered



VI. User testing

Feedback was gathered from a participant, a 30-year-old parent of a 4-year-old child, during testing of the app prototype. The full list of questions can be found in the Appendix. Overall, the participant was able to navigate the main features and complete all core tasks; however, several usability issues emerged during interaction that affected efficiency and clarity.

One key issue involved the main screen, where the participant did not initially recognize that additional content was available through scrolling. As a result, important information was overlooked until prompted, suggesting that the scrollable area lacked sufficient visual affordances and did not clearly communicate its interactive nature. In the Story section, the participant hesitated when attempting to play audio content and noted that the placement of the playback button was not immediately intuitive, requiring additional time to locate and understand its function. Another issue arose within the real-time intercom feature. While the participant was able to access the function, he reported uncertainty about which device was currently connected, as no clear status indicator was present on the interface. This lack of system feedback reduced confidence in using the feature, particularly in a context where reliability and clarity are important for parent-child communication.

Despite these issues, the participant was able to successfully complete all tasks and reported a generally positive experience, suggesting that the overall structure of the interface is understandable, though certain elements require improved clarity and feedback to better support user interaction.

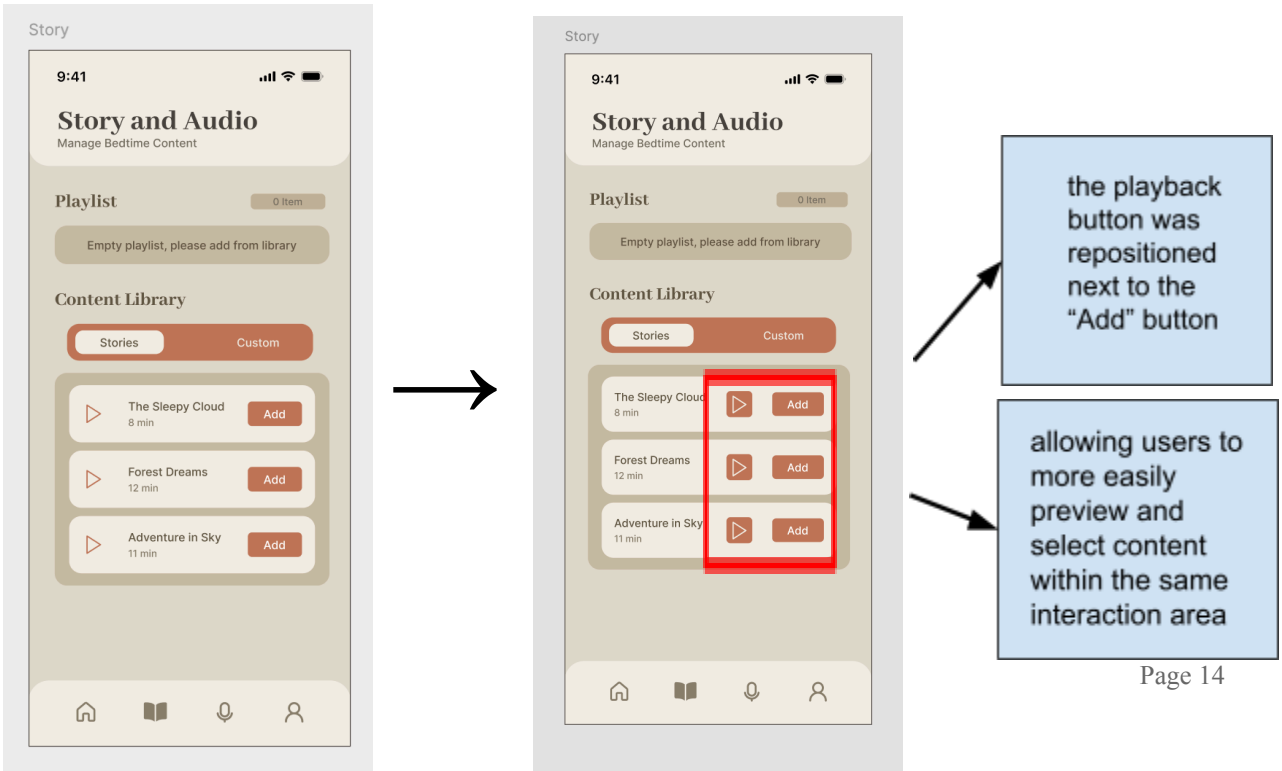
VII. App Interface Redesign

The Lumino APP interface has been redesigned after the user testing.

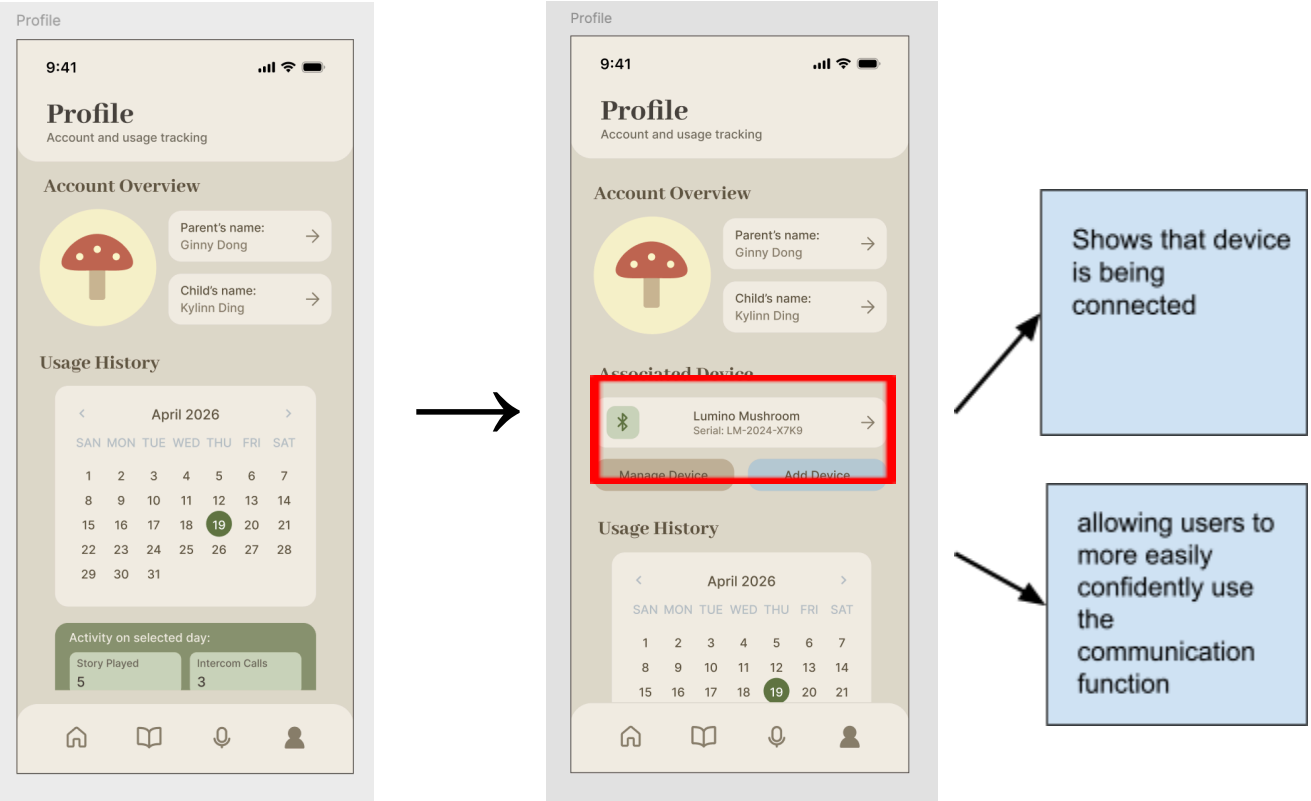
Redesign 1:



Redesign 2:



Redesign 3:



VIII. Discussion and reflection

This project provided a full experience of the product development process, from research to design and user testing. Through methods learned in ENP 64 Methods of Human Factors Engineering, including CLIMBER analysis, questionnaires, and task analysis, a clearer understanding was developed of the challenges both children and caregivers face during the transition to independent sleep. These methods revealed that the process involves not only physical routines, but also emotional and cognitive demands, including repeated cycles of uncertainty and reassurance. This insight directly informed the design of Lumino as a system that supports both users.

Applying Human Factors principles such as visibility, feedback, and reducing cognitive load showed how small design decisions can significantly impact usability. User testing highlighted issues in clarity and interaction, which were addressed through iterative redesigns that improved affordances, control placement, and system feedback. This reinforced the importance of iteration in creating effective designs.

Overall, this project demonstrated how the methods and principles from ENP 64 can be applied in a real-world context to develop user-centered solutions. Future work could include additional testing with a broader user group and further refinement of the integration between the physical device and the app.

IX. Appendix

Questionnaire content

The following questionnaire was designed to understand bedtime routines, sleep challenges, and caregiver responses for children ages 1–5 during the transition to independent sleep.

1. How old is your child?
   1–2
   2–3
   3–4
   4–5
2. Is your child currently transitioning to sleeping independently?
   Yes
   No
3. How often does your child have difficulty falling asleep independently?
   Never
   Rarely
   Sometimes
   Often
   Always
4. Does your child wake up during the night?
   No
   Yes (1–2 times)
   Yes (3+ times)
5. When your child calls out at night, how do you usually respond?
   Immediately go to their room
   Wait a few minutes before responding

Respond verbally from another room

Encourage them to self-soothe

6. How many times do you typically re-enter your child's room during bedtime?

   0

   1–2

   3–5

   5+

7. How stressful do you find your child's bedtime routine?

   Not stressful

   Slightly stressful

   Moderately stressful

   Very stressful

8. What are some ways that you are currently using to help your child fall asleep?
9. Does your child currently use any interactive sleep aids (devices, apps, etc.)?

   Yes

   No

10. How comfortable would you be with a device that allows your child to communicate with you without leaving bed?

    Not comfortable

    Slightly comfortable

    Neutral

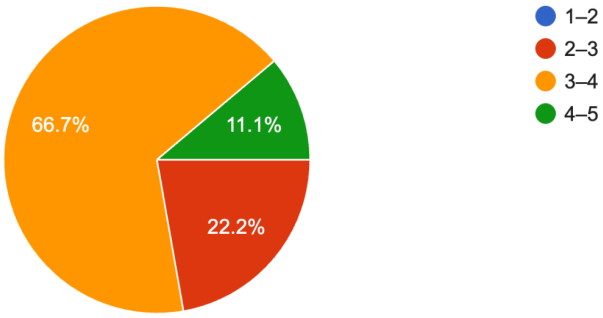
    Comfortable

    Very comfortable

Questionnaire results

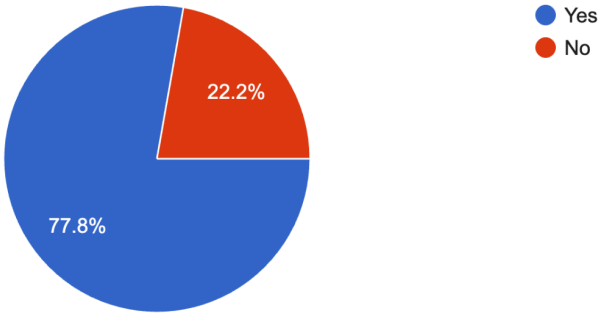
How old is your child?

9 responses



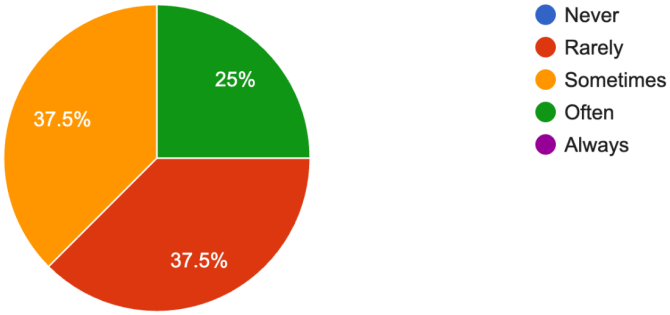
Is your child currently transitioning to sleeping independently?

9 responses



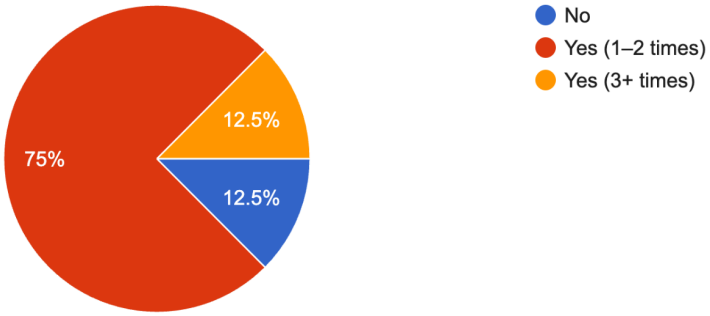
How often does your child have difficulty falling asleep independently?

8 responses



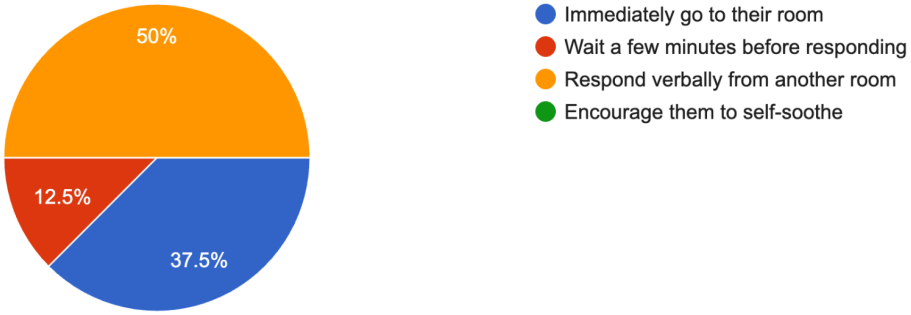
Does your child wake up during the night?

8 responses



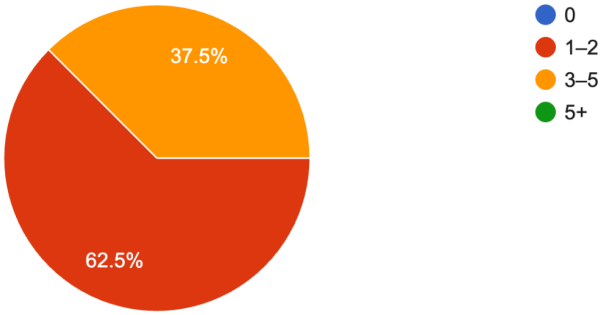
When your child calls out at night, how do you usually respond?

8 responses



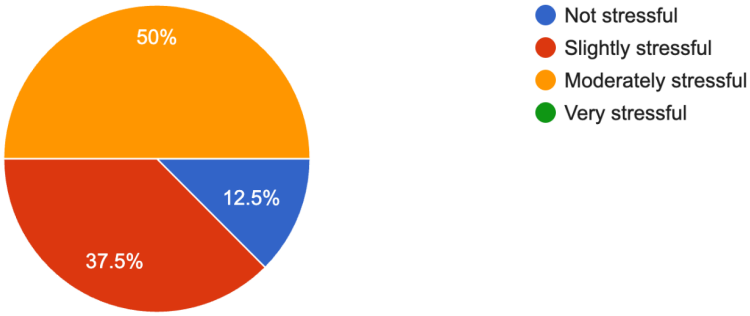
How many times do you typically re-enter your child's room during bedtime?

8 responses



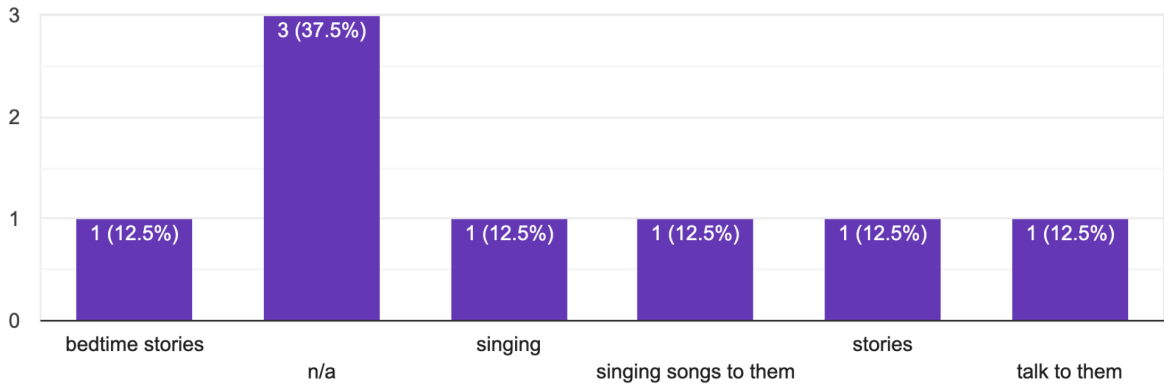
How stressful do you find your child's bedtime routine?

8 responses



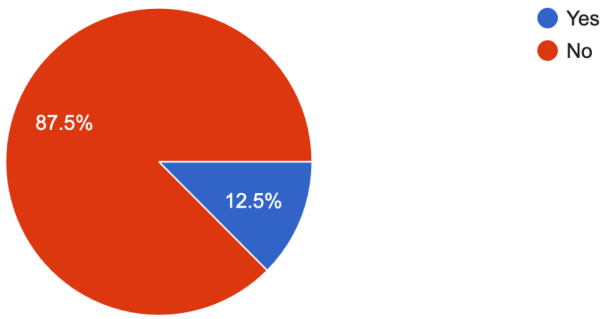
What are some ways that you are currently using to help your child fall asleep?

8 responses



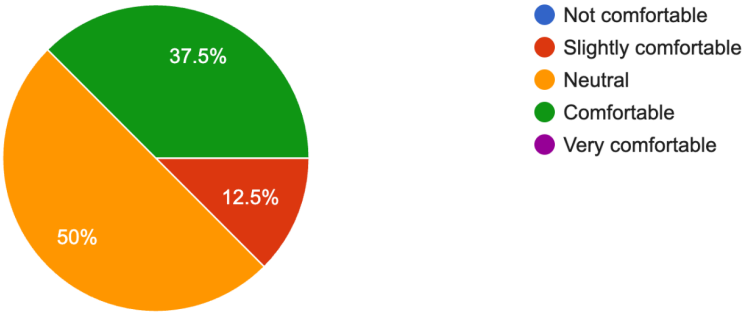
Does your child currently use any interactive sleep aids (devices, apps, etc.)?

8 responses

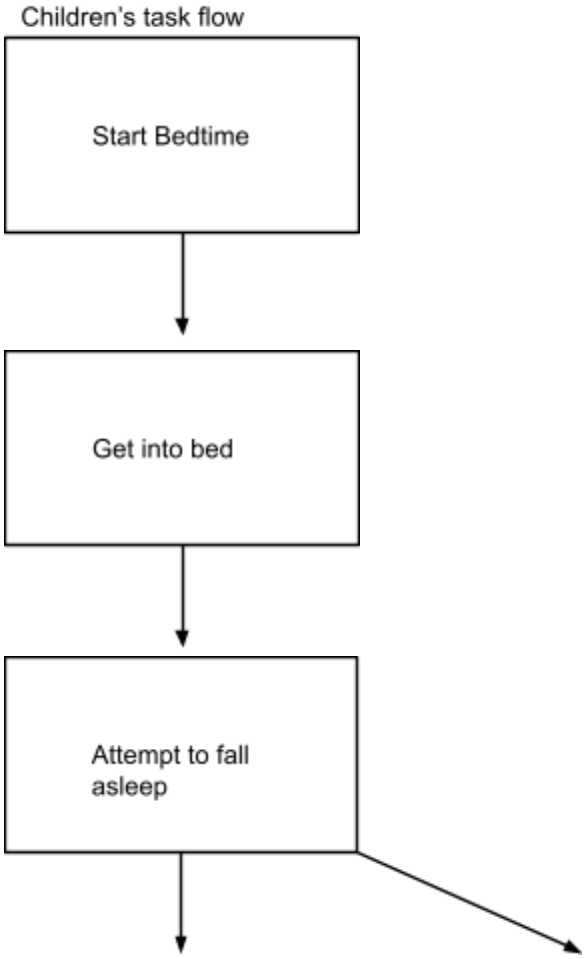


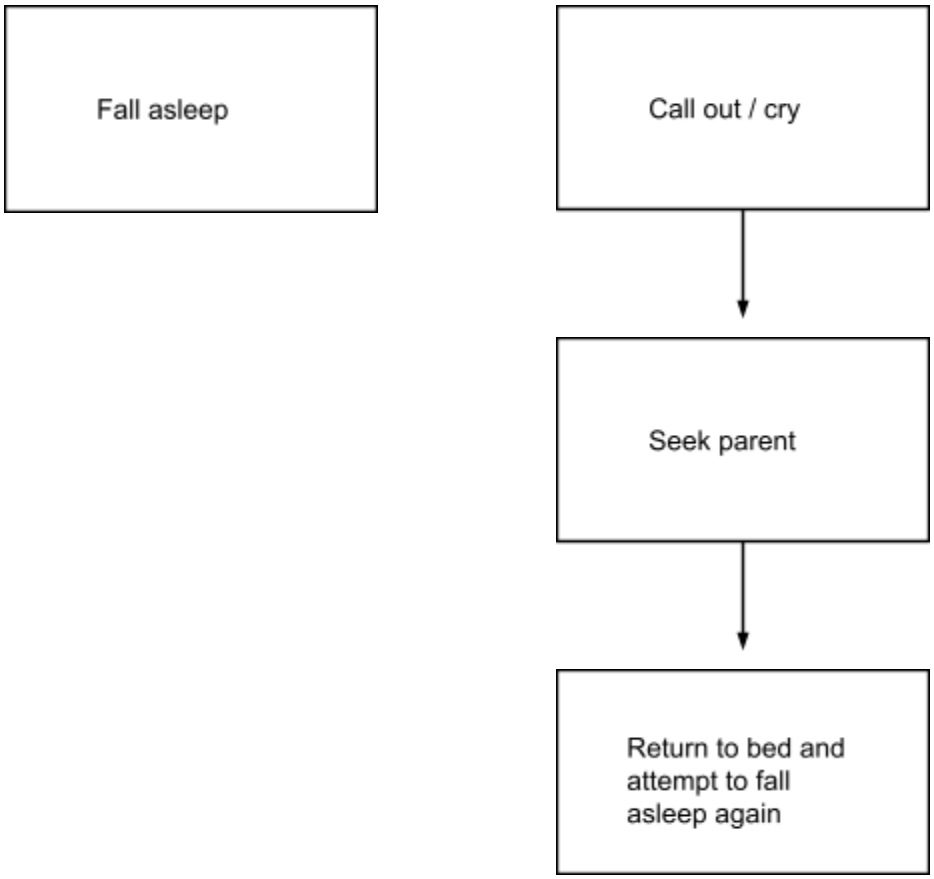
How comfortable would you be with a device that allows your child to communicate with you without leaving bed?

8 responses

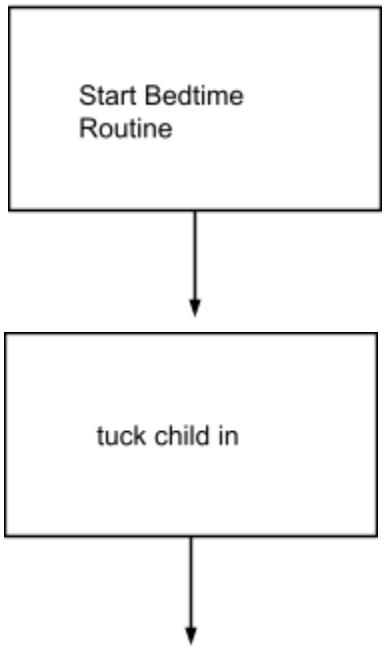


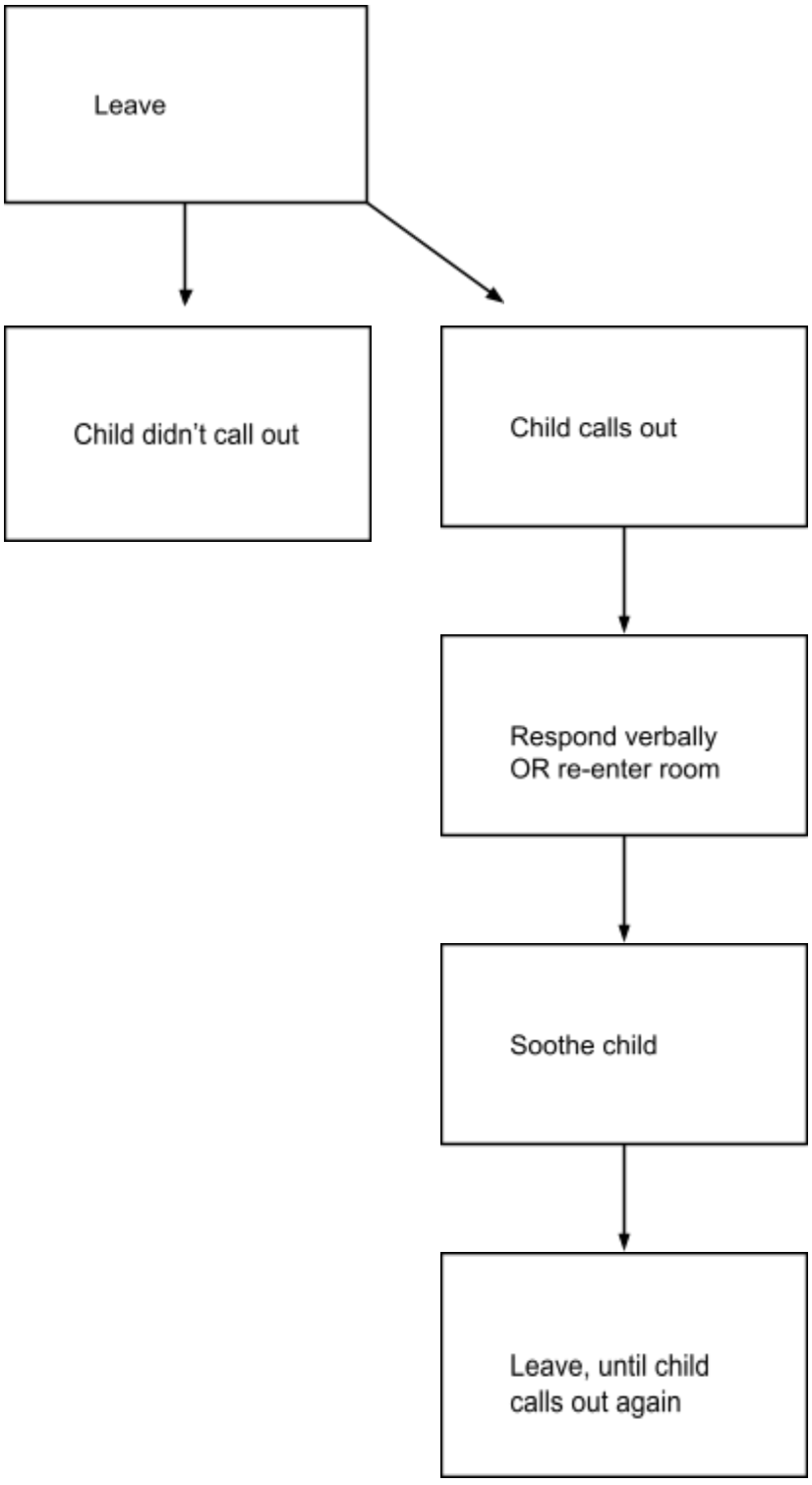
Task analysis





Parent's task flow




Leave
Child didn't call out
Child calls out
Respond verbally
OR re-enter room
Soothe child
Leave, until child
calls out again

Usability testing script

Thank you for joining us today. We're conducting a usability test for Lumino, a companion app for a children's nightlight device designed for parents of young children.

I want to emphasize that we're testing the interface, not you. There are no right or wrong answers. Any feedback you have, positive or critical, is exactly what we're looking for.

This session will be recorded for the purpose of this study only. Your responses will remain confidential and will only be reviewed by our team. You may pause or stop at any time. Is it okay if I also take notes during our session?

Background Questions:

1. Could you tell me your age and your role — are you a parent, caregiver, or something else?
2. How old are your children?
3. How would you describe your bedtime routine with your child right now?
4. How comfortable are you with using apps on a daily basis?

Task 1: Please make your way to the home screen however feels natural to you, and adjust the settings of the light. Was it easy to navigate?

Task 2: Now imagine you want your child to hear a story tonight. Please try to find the section that allows you to add stories to playlist. Was it easy to navigate?

Task 3: Imagine your child is in bed and you want to check in on them. Please try to use the intercom feature to speak to them. Was it easy to navigate?

Task 4: Please navigate to the profile page. Is it clear? Do you think any section is missing from the profile page?

SUS-Style Rating Scale: I'll read a statement and please respond with: Strongly Disagree / Somewhat Disagree / Neutral / Somewhat Agree / Strongly Agree

1. I found the app unnecessarily complex.

2. I felt confident navigating the app.
3. I would need technical support to use this app.
4. The different features of the app felt well integrated.
5. I think most parents would learn to use this app quickly.
6. The main purpose of the app was clear to me.

Close

Thank you so much for your time. Your feedback is incredibly valuable and will directly shape the final product.